

Two Variable Systems Practice

Solve each system of equations by substitution or elimination. Remember to indicate one solution, no solution or infinitely many solutions.

1. $x + 2y = 13$

$3x - 5y = 6$

$x = -2y + 13$

$3(-2y + 13) - 5y = 6$

$-6y + 39 - 5y = 6$

$-11y + 39 = 6$

$-11y = -33$

$y = 3$

$x + 2(3) = 13$

$x = 7$

$(7, 3)$

2. $-4c - 2d = -2$

$-1(2c - 2d = -14)$

$-4c - 2d = -2$

$-2c + 2d = 14$

$-6c = 12$

$c = -2$

$2(-2) - 2d = -14$

$-4 - 2d = -14$

$+4 \quad +4$

$-2d = -10$

$d = 5$

$(-2, 5)$

3. $x + 5y = 4$

$3x + 15y = -1$

$x = -5y + 4$

$3(-5y + 4) + 15y = -1$

$-15y + 12 + 15y = -1$

$12 = -1$

No soln.

4. $5x - 2y = 14$

$-2(2x - y = 5)$

$5x - 2y = 14$

$-4x + 2y = -10$

$x = 4$

$2(4) - y = 5$

$8 - y = 5$

$-y = -3$

$y = 3$

$(4, 3)$

5. $6m - 8n = 3$

$-1(2m - 8n = -3)$

$6m - 8n = 3$

$-2m + 8n = 3$

$4m = 6$

$m = \frac{6}{4} = \frac{3}{2}$

$2(\frac{3}{2}) - 8n = -3$

$3 - 8n = -3$

$-8n = -6$

$-8 \quad -8$

$n = \frac{3}{4}$

$(\frac{3}{2}, \frac{3}{4})$

6. $-\frac{1}{3}x - \frac{4}{3}y = -2$

$\frac{1}{3}x - \frac{2}{3}y = 4$

$-2y = 2$

$y = -1$

$\frac{1}{3}x + \frac{2}{3} = 4$

$x + 2 = 12$

$x = 10$

$(10, -1)$

7. $3x - y = 4$

$2x - 3y = -9$

$-9x + 3y = -12$

$2x - 3y = -9$

$-7x = -21$

$-7 \quad -7$

$x = 3$

$4 - y = 4$

$-y = 0$

$y = 0$

$(3, 0)$

8. $x - 5y = 10$

$2x - 10y = 20$

$-2x + 10y = -20$

$2x - 10y = 20$

$0 = 0$

inf. many sol.

$$9. \begin{cases} 3x - 4y = 8 \\ 6x - 8y = 9 \end{cases}$$

$$\begin{array}{r} 6x - 8y = 9 \\ -6x + 8y = -16 \\ \hline 0 = -7 \end{array}$$

No soln

$$10. \begin{cases} 3x - 8y = -6 \\ -5x + 4y = 10 \end{cases}$$

$$\begin{array}{r} 3x - 8y = -6 \\ -10x + 8y = 20 \\ \hline -7x = 14 \end{array}$$

$$x = -2$$

$$3(-2) - 8y = -6$$

$$-6 - 8y = -6$$

$$-8y = 0$$

$$y = 0$$

$(-2, 0)$

$$11. \begin{cases} 4x - 5y = 17 \\ 3x + 4y = 5 \end{cases}$$

$$\begin{array}{r} 12x - 15y = 51 \\ -12x - 16y = -20 \\ \hline -31y = 31 \end{array}$$

$$y = -1$$

$$\begin{array}{r} 3x - 4 = 5 \\ 3x = 9 \\ x = 3 \end{array}$$

$(3, -1)$

$$12. \begin{cases} 2c + 6d = 14 \\ -\frac{7}{3} + \frac{1}{3}c = -d \end{cases}$$

$$\begin{array}{r} -\frac{7}{3} + \frac{1}{3}c = -d \\ -6\left(\frac{1}{3}c + d = \frac{7}{3}\right) \end{array}$$

$$\begin{array}{r} 2c + 6d = 14 \\ -2c - 6d = -14 \\ \hline 0 = 0 \end{array}$$

inf. many sol

13. All 28 members in Crestview High School's Ski Club went on a one-day ski trip. Members can rent skis for \$16 per day or snowboards for \$19 per day. The club paid a total of \$478 for rental equipment. Write a system of equations that represents the number of members who rented the two types of equipment. How many members rented skis and how many rented snowboards?

$$\begin{array}{l} \text{skis} = x \\ \text{boards} = y \end{array}$$

$$\begin{cases} x + y = 28 \\ 16x + 19y = 478 \end{cases}$$

$$x = 28 - y$$

$$16(28 - y) + 19y = 478$$

$$448 - 16y + 19y = 478$$

$$448 + 3y = 478$$

$$3y = 30$$

$$\begin{array}{l} y = 10 \text{ boards} \\ x = 18 \text{ skis} \end{array}$$

14. Tickets to a movie cost \$7.25 for adults and \$5.50 for students. A group of friends purchased 8 tickets for \$52.75. Write a system of equations to represent the situation. How many adult tickets and student tickets were purchased?

$$\begin{cases} x + y = 8 \\ 7.25x + 5.5y = 52.75 \end{cases}$$

$$x = 8 - y$$

$$7.25(8 - y) + 5.5y = 52.75$$

$$58 - 7.25y + 5.5y = 52.75$$

$$-1.75y = -5.25$$

$$y = 3$$

3 student
5 adult